

Week 10: Causal Inference, Foundation Models and more

Machine Learning and Advanced Analytics — Exam Notes

These notes are based mainly on **Causal Inference 2025** and the neural architecture / transformers material from **ML_NNArchitectures**.

1. Why causal inference matters

So far, most machine learning models we have used are **association / observational models**.

A normal predictive model learns:

$$P(y \mid x, z)$$

This means:

Given the data we observed, how are inputs related to the output?

Example:

A churn model may learn that customers who do not use a data plan are more likely to churn.

But this does **not automatically mean** that forcing customers onto a data plan will reduce churn.

Causal inference asks a different question:

$$P(y \mid do(x = 1), z)$$

This means:

What would happen to the outcome if we actively intervened and changed x ?

So the key difference is:

Predictive ML	Causal Inference
Learns patterns in observed data	Estimates effect of an intervention
Useful for prediction	Useful for action and decision-making
Answers "who is likely to churn?"	Answers "what should we change to reduce churn?"
Based on association	Based on cause-effect reasoning

2. Association vs causation

A variable can be important in a machine learning model without being causal.

For example:

A model may find that **ice cream sales** and **drowning incidents** are associated.

But ice cream does not cause drowning. A hidden factor, such as **hot weather**, may influence both.

This is why causal inference matters in business analytics:

Predictive models tell us what is related to the outcome.

Causal models help us decide what action will actually change the outcome.

In the churn case study, predictive modelling can identify customers likely to churn, but causal reasoning is needed to decide whether an intervention such as a voucher, data plan, offer, or campaign will actually reduce churn.

3. What is causality?

A treatment T causes an outcome Y if changing T leads to a change in Y , while keeping everything else constant.

The **causal effect** is the size of the change in Y caused by a unit change in T .

Example:

If sending a coupon increases the probability of purchase from 20% to 30%, then the causal effect is:

$$30\% - 20\% = 10\%$$

Here:

Term	Meaning
Treatment T	The intervention, e.g. coupon sent or not sent
Outcome Y	The result, e.g. purchase or no purchase
Causal effect	Difference in outcome caused by the treatment
Counterfactual	What would have happened under the alternative treatment

4. Counterfactual thinking

Causal inference is based on **counterfactuals**.

A counterfactual asks:

What would have happened to the same person/customer if they had received a different treatment?

Example:

A customer receives a coupon and buys.

Observed world:

$$Y_{T=1} = 1$$

Counterfactual world:

$$Y_{T=0} = ?$$

We do not know whether the customer would have bought without the coupon.

This creates the **fundamental problem of causal inference**:

For each individual, we can only observe one outcome: either the treated outcome or the untreated outcome. We cannot observe both for the same person at the same time.

5. Potential outcomes framework

The potential outcomes framework represents the two possible outcomes for each individual:

$$Y_{T=1}$$

Outcome if treated.

$$Y_{T=0}$$

Outcome if not treated.

The individual treatment effect is:

$$Y_{T=1} - Y_{T=0}$$

But because we cannot observe both outcomes for the same person, causal inference tries to estimate the missing counterfactual.

Example:

Person	Treatment observed	Buy if treated	Buy if untreated
P1	0	?	0
P2	1	1	?
P3	1	1	?

We observe only one side for each person. The other side is missing.

6. Average Treatment Effect, ATE

The **Average Treatment Effect**, or **ATE**, measures the average causal effect across the whole population.

$$ATE = E[Y_{T=1} - Y_{T=0}]$$

In simple words:

What is the average effect of giving the treatment to everyone compared with giving it to no one?

Business example:

A company wants to know whether sending a voucher increases purchases across all customers.

ATE answers:

On average, would customers buy more if everyone received the voucher?

This is usually what analysts want when they want a general policy decision.

7. Average Treatment Effect on the Treated, ATT

The **Average Treatment Effect on the Treated**, or **ATT**, measures the average causal effect only for the people who actually received the treatment.

$$ATT = E[Y_{T=1} - Y_{T=0} \mid T = 1]$$

In simple words:

What was the effect of the treatment on the people who were treated?

Example:

If only high-value customers received vouchers, ATT tells us the effect for those high-value customers.

But it may not generalise to all customers.

Measure	Meaning	Generalises to
ATE	Average effect across the whole population	Everyone
ATT	Average effect among treated units only	People similar to those treated

Important exam point:

ATE and ATT are not always the same, especially when treatment is not randomly assigned.

8. Why averages can hide important effects

Average effects can hide subgroup differences.

Example:

Suppose a voucher:

Group	Effect
Under 18 customers	Doubles probability of buying
18+ customers	Halves probability of buying

If the population is 50% under 18 and 50% 18+, the average effect may look like zero.

But from a business perspective, the intervention is clearly useful for one group and harmful for another.

Exam interpretation:

Averages are useful summaries, but they can hide heterogeneity. Analysts should consider whether treatment effects differ across customer groups.

9. Randomised experiments: the gold standard

The best way to estimate causal effects is a **randomised experiment**.

In a randomised experiment, people are randomly assigned to:

Group	Meaning
Treatment group	Receives the intervention
Control group	Does not receive the intervention

Because assignment is random, confounders should be balanced across both groups on average.

So differences in outcomes can be attributed to the treatment.

Example:

A retailer randomly sends a voucher to 50% of customers and does not send it to the other 50%.

If the treated group buys more, and the groups were properly randomised, we can argue the voucher caused the increase.

10. A/B testing

A/B testing is a practical form of randomised experiment used in business.

Definition:

A/B testing is controlled statistical testing of an intervention in the real world to determine whether it has a positive, negative, or no effect.

Examples:

Context	A/B test
Website	Old layout vs new layout
Marketing	No email vs promotional email
Retail	Normal shelf layout vs new display
Product	Existing feature vs redesigned feature

A/B testing is powerful because it directly tests an intervention.

However, it can be:

Limitation	Explanation
Costly	Requires time, data, and operational setup
Risky	A bad intervention may reduce sales
Unethical	It may be wrong to deny a useful treatment
Hard to implement	Randomisation may fail in real-world systems

11. Confounding

A **confounder** is a variable that affects both the treatment and the outcome.

Example:

Suppose we test whether a new website algorithm increases purchases.

A hidden variable, such as income, may affect:

1. Which algorithm the customer receives
2. Whether the customer buys

Then income is a confounder.

The causal diagram is:

$$\begin{array}{l} \text{Income} \rightarrow \text{Algorithm} \\ \text{Income} \rightarrow \text{Purchase} \end{array}$$

This creates a false impression that the algorithm caused the change in purchases.

12. Simpson's paradox

Simpson's paradox happens when the overall trend differs from the trend within subgroups.

From the lecture example:

Group	Old algorithm A	New algorithm B
Low-income visitors	$10/400 = 2.5\%$	$4/200 = 2.0\%$
High-income visitors	$40/600 = 6.6\%$	$50/800 = 6.2\%$
Overall	$50/1000 = 5.0\%$	$54/1000 = 5.4\%$

Overall, the new algorithm looks better:

$$5.4\% > 5.0\%$$

But within both subgroups, the new algorithm is worse:

$$\begin{array}{l} 2.0\% < 2.5\% \\ 6.2\% < 6.6\% \end{array}$$

So the new algorithm only looks better because more high-income visitors were exposed to it.

Exam interpretation:

Simpson's paradox shows that aggregate statistics can be misleading when treatment groups differ in composition. If randomisation fails or confounders are not controlled, we may reach the wrong causal conclusion.

13. Causal inference workflow in business

A typical business analytics workflow is:

Step 1: Build a predictive model

Use machine learning to predict an outcome such as churn.

Example:

$$P(\text{churn} \mid \text{customer features})$$

This identifies customers at risk.

Step 2: Understand important predictors

Use variable importance, profiles, or pen portraits.

Example:

Customers with low engagement and no data plan may be more likely to churn.

Step 3: Identify possible interventions

Ask:

Which important variables can the business actually change?

Example:

A business cannot directly change customer age, but it can change:

- Voucher offers
- Email frequency
- Subscription plan
- Campaign style
- Product placement
- Price discount

Step 4: Estimate causal effect

Check whether changing the variable would actually change the outcome.

Example:

Does offering a data plan reduce churn, or is data-plan usage only associated with lower churn?

Step 5: Target intervention

Use the predictive model to decide who should receive the intervention.

The best target customers are not always the customers most likely to churn.

A good target is someone who is likely to change behaviour because of the intervention.

14. Why neural architecture matters

A fully connected network can theoretically approximate any continuous function, but this does not mean it is practical.

The Universal Approximation Theorem says that a shallow neural network with enough neurons can approximate any continuous function.

But the problem is:

The theorem says the correct weights exist. It does not guarantee gradient descent will find them efficiently with limited data and compute.

So different neural architectures exist because they build useful assumptions into the model.

These assumptions are called **inductive biases**.

15. Inductive bias

An **inductive bias** is an assumption built into a learning algorithm to help it generalise to unseen data.

Fully connected networks have weak inductive bias.

They assume:

Any input feature can directly interact with any other input feature.

This is useful for tabular data, but inefficient for structured data such as images, text, audio, or time series.

Examples:

Data type	Useful architecture	Inductive bias
Tabular data	Fully connected network	Features treated as general inputs
Images	CNN	Nearby pixels form meaningful patterns
Time series	RNN / LSTM	Order and memory matter
Text	Transformer	Words attend to relevant context

16. Why fully connected networks fail on images

Fully connected networks can perform badly on images because of three main issues.

1. Parameter explosion

A 200×200 colour image has:

$$200 \times 200 \times 3 = 120,000$$

input values.

If the first hidden layer has 1,000 neurons:

$$120,000 \times 1,000 = 120,000,000$$

weights in the first layer alone.

This is inefficient and likely to overfit.

2. Spatial blindness

Images have spatial structure.

Nearby pixels form edges, textures, and objects.

But a fully connected network flattens the image into a one-dimensional vector, destroying the 2D structure.

3. Lack of translation invariance

If a cat appears on the left during training but on the right during testing, a fully connected network may fail because it learned location-specific patterns.

It may memorise where the object appeared rather than what the object is.

17. Convolutional Neural Networks, CNNs

CNNs solve the problems of fully connected networks on images by using three principles.

1. Local receptive fields

A neuron looks only at a small local patch of the image, such as 3×3 pixels.

This preserves spatial structure.

2. Weight sharing

The same filter slides across the whole image.

This means the model can detect the same feature anywhere in the image.

Example:

A horizontal edge detector is useful in the top-left, centre, or bottom-right of the image.

3. Pooling

Pooling compresses feature maps and makes the model less sensitive to exact position.

Example:

Max pooling keeps the strongest signal in a small region.

This helps with translation invariance.

Summary:

FCN problem	CNN solution
Parameter explosion	Weight sharing
Spatial blindness	Local receptive fields
Object location sensitivity	Pooling

18. Neural architectures for temporal data

Some data is ordered.

Examples:

- Speech
- Text
- Video
- Financial data
- Customer purchase sequences
- Website clickstream data

For temporal problems, the order of events matters.

A fully connected network or standard CNN may treat inputs as independent, but temporal models need memory.

19. Recurrent Neural Networks, RNNs

RNNs process data step by step.

At time t , the hidden state depends on:

1. The current input
2. The previous hidden state

So the model has memory.

Simple idea:

$$h_t = f(x_t, h_{t-1})$$

Where:

Symbol	Meaning
x_t	Input at time t
h_t	Hidden state at time t
h_{t-1}	Memory from previous time step

RNNs are useful because:

Feature	Meaning
Persistence	Past information can influence future predictions
Weight sharing	Same network block is reused across time
Sequence handling	Can process ordered data

Example:

A customer's earlier purchases may affect whether they churn later.

20. Limitation of RNNs

Simple RNNs struggle with long-term dependencies.

Information from early time steps must pass through many intermediate steps.

This can cause the model to forget earlier information.

Example:

In a long customer history, an important purchase event from many months ago may be lost by the time the model predicts churn.

This is called the **short-term memory problem**.

21. Long Short-Term Memory, LSTM

LSTMs are a type of RNN designed to handle longer-term dependencies.

They use two memory pathways:

Memory type	Meaning
Short-term memory	Changes heavily at each step

Memory type	Meaning
Long-term memory	Carries important information across many steps

LSTMs use **gates** to control information flow.

Important gates:

Gate	Function
Forget gate	Decides what old information to remove
Input gate	Decides what new information to store
Output gate	Decides what information to output

LSTMs are useful when earlier events may matter much later.

Example:

In text, the subject at the start of a sentence may affect the meaning of a later word.

22. Encoder-decoder models

Encoder-decoder models are used for sequence-to-sequence tasks.

Example:

Input	Output
Sentence in French	Sentence in English
Image	Caption
Video segment	Description

The encoder reads the input sequence and compresses it into a **context vector**.

The decoder uses that context vector to generate the output sequence.

Problem:

A single fixed-size context vector becomes a bottleneck for long sequences.

It is like trying to summarise a long book on one sticky note.

23. Attention

Attention solves the encoder-decoder bottleneck.

Instead of compressing the whole input into one fixed-size vector, attention allows the decoder to look back at different parts of the input when generating each output.

Simple explanation:

Attention lets the model focus on the most relevant parts of the input at each step.

Example:

When translating a sentence, the model does not rely only on a single summary. It can look back at the relevant word or phrase in the original sentence.

This restores more direct influence between distant parts of a sequence.

24. Transformers

Transformers are based on attention, especially **self-attention**.

Old attention:

When generating output, look back at encoder states and decide which input parts are relevant.

Self-attention:

For each word, decide which other words in the same sentence are relevant.

The key advantage is that self-attention can be computed in parallel.

This makes transformers much faster to train than RNNs or LSTMs on modern GPUs.

25. Token embeddings and positional encodings

Transformers cannot process raw text directly.

Text is converted into numerical representations.

Token embeddings

Words or subword tokens are converted into vectors.

Example:

"customer" → numerical vector

Positional encodings

Because transformers process tokens in parallel, they need information about word order.

So positional encodings are added.

$$\text{Final Input} = \text{Token Embedding} + \text{Positional Encoding}$$

This lets the model know both:

1. What the token is
 2. Where the token appears in the sequence
-

26. Multi-head self-attention

Transformers use multiple attention heads.

Each head can learn a different type of relationship.

Example:

One head may focus on grammar.

Another may focus on long-range meaning.

Another may focus on nearby words.

The outputs of these heads are combined to produce a richer representation.

Simple explanation:

Multi-head attention is like having multiple specialists looking at the same sentence from different perspectives.

27. Foundation models

Foundation models are large models trained on very large datasets and then adapted to many downstream tasks.

Examples include:

- Large language models
- Vision-language models
- Code models
- Text embedding models

They are called “foundation” models because they act as a base for many different applications.

Tasks include:

Task	Example
Text generation	Writing summaries
Classification	Sentiment analysis
Question answering	Answering based on documents
Translation	English to French
Coding	Generating Python code
Embeddings	Semantic search

28. Why transformers are the basis of foundation models

Transformers are useful for foundation models because they:

Advantage	Explanation
Handle long context	Self-attention connects tokens directly
Train efficiently	Parallel computation on GPUs
Scale well	Larger models and datasets improve capability
Transfer well	Pre-trained knowledge can be adapted to many tasks
Learn contextual meaning	Same word can have different meaning depending on context

This is why modern systems such as ChatGPT are based on transformer-style architectures.

29. Link between causal inference and foundation models

Causal inference and foundation models answer different kinds of questions.

Topic	Main question
Predictive ML	What is likely to happen?
Causal inference	What will happen if we intervene?
Foundation models	How can large pre-trained models understand/generate/adapt across tasks?

Important distinction:

A foundation model may be very good at prediction and pattern recognition, but that does not automatically mean it understands causality.

Example:

An LLM may know that customers who receive discounts often buy more.

But causal inference is needed to answer:

Did the discount cause the purchase, or were likely buyers targeted with discounts?

So foundation models can support business analytics, but causal reasoning is still needed for action.

30. Exam-ready comparison table

Concept	Key idea	Exam phrase
Association	Relationship in observed data	Correlation does not imply causation
Causality	Changing treatment changes outcome	Requires counterfactual reasoning
Counterfactual	What would have happened otherwise	Cannot observe both outcomes for same unit
ATE	Average effect in population	Useful for general policy decisions
ATT	Average effect among treated group	May not generalise to whole population
Randomised experiment	Random treatment assignment	Gold standard for causal inference
A/B test	Business experiment in the wild	Tests intervention effect
Confounder	Affects treatment and outcome	Can bias causal estimates
Simpson's paradox	Overall trend differs from subgroup trends	Aggregates can mislead
Inductive bias	Built-in modelling assumption	Helps learning/generalisation
CNN	Local image structure	Uses local receptive fields, weight sharing, pooling
RNN	Sequential memory	Uses previous hidden state
LSTM	Longer-term sequence memory	Uses gates
Attention	Selective focus	Looks at relevant input parts
Transformer	Self-attention architecture	Foundation of modern LLMs
Foundation model	Large pre-trained adaptable model	Base model for many downstream tasks

31. Likely exam-style questions and answer points

Q1. Explain the difference between predictive modelling and causal inference.

Predictive modelling learns associations in observed data, usually estimating $P(y|x, z)$. It can predict outcomes but does not tell us whether changing a variable will change the outcome. Causal inference estimates what would happen under an intervention, using ideas such as $P(y|do(x), z)$. It is needed for decision-making because business actions require knowing whether an intervention causes a change.

Q2. What is the fundamental problem of causal inference?

For any individual, we can only observe one potential outcome: either the outcome under treatment or the outcome without treatment. We cannot observe both for the same person at the same time. Therefore, causal inference is the problem of estimating the missing counterfactual.

Q3. Why are randomised experiments considered the gold standard?

Randomised experiments randomly assign units to treatment and control groups. This should balance both observed and unobserved confounders across groups on average. Therefore, differences in outcomes can be attributed to the treatment rather than pre-existing group differences.

Q4. Explain Simpson's paradox using the algorithm example.

The new algorithm appears better overall because it has a higher total conversion rate. However, within both low-income and high-income groups, the old algorithm performs better. The overall result is misleading because the new algorithm was shown to more high-income visitors, who are more likely to buy anyway. This shows that group composition and confounding can reverse conclusions.

Q5. Why do we need architectures like CNNs if fully connected networks are universal approximators?

Universal approximation says the right weights exist, but it does not mean gradient descent can find them efficiently. Fully connected networks ignore structure in data, causing parameter explosion, overfitting, and inefficient learning. CNNs add inductive bias for images through local receptive fields, weight sharing, and pooling, making learning more efficient and generalisable.

Q6. Why are transformers better suited than RNNs for foundation models?

RNNs process sequences step by step, which makes them slow and causes difficulty with long-range dependencies. Transformers use self-attention, allowing each token to directly attend to

other relevant tokens. This can be computed in parallel, scales better on GPUs, and supports large-scale pre-training, making transformers suitable for foundation models.

32. One-page memory summary

Week 10 is mainly about moving from **prediction to action**.

Predictive ML finds associations, but causal inference asks what happens if we intervene. Causality is based on counterfactual reasoning: comparing what happened with what would have happened under another treatment. The fundamental problem is that we cannot observe both outcomes for the same individual. Average Treatment Effect measures the average effect across the population, while ATT measures the effect only for treated units. Randomised experiments and A/B tests are the gold standard because randomisation balances confounders. If randomisation fails, confounding and Simpson's paradox can produce misleading conclusions.

The second part explains why neural architectures matter. Fully connected networks can theoretically approximate any function, but they are inefficient for structured data. CNNs encode spatial assumptions for images. RNNs and LSTMs encode sequence and memory assumptions for temporal data. Attention improves sequence modelling by letting models focus on relevant parts of the input. Transformers use self-attention and parallel computation, making them the basis of modern foundation models.